The Let d be the distance between the double slits S_1 and S_2 which act as coherent sources of wavelength λ. A screen is placed parallel to the double slit at a distance D from it. The mid-point of S_1 and S_2 is C and the mid-point of the screen O is equidistant from S_1 and S_2. P is any point at a distance y from O. The waves from S_1 and S_2 meet at P either in-phase or out-of-phase depending upon the path difference between the two waves.

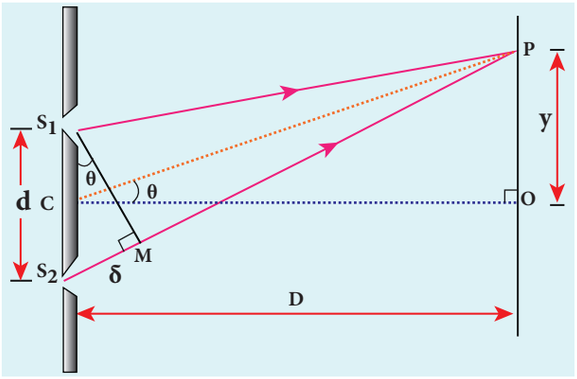


Figure 7.13 Young's double slit arrangement to find path difference

The path difference δ between the light waves from S_1 and S_2 to the point P is, $\delta = S_2P - S_1P$

A perpendicular is dropped from the point S_1 to the line S_2P at M to find the path difference more precisely.

$$\delta = S_2P - MP = S_2M \qquad (7.26)$$

The angular position of the point P from C is θ. $\angle OCP = \theta$.

From the geometry, the angles $\angle OCP$ and $\angle S_2S_1M$ are equal.

$\angle OCP = \angle S_2S_1M = \theta$.

In right angle triangle ΔS_1S_2M, the path difference, $S_2M = d \sin\theta$

$$\delta = d \sin\theta \qquad (7.27)$$

If the angle θ is small, $\sin\theta \approx \tan\theta \approx \theta$

From the right angle triangle ΔOCP, $\tan\theta = \frac{y}{D}$

$$\text{The path difference, } \delta = \frac{d\,y}{D} \qquad (7.28)$$

Based on the condition of the path difference, the point P may have a bright (or) dark fringe.

Condition for bright fringe (or) maxima

The condition for the point P to have a constructive interference (or) be a bright fringe is,

Path difference, $\delta = n\lambda$

where, $n = 0, 1, 2, \ldots$

$$\therefore \frac{d\,y}{D} = n\lambda$$

$$y = n\frac{\lambda D}{d} \quad \text{(or)} \quad y_n = n\frac{\lambda D}{d} \qquad (7.29)$$

This is the condition for the point P to have a bright fringe. The distance y_n is the distance of the nth bright fringe from the point O.

Condition for dark fringe (or) minima

The condition for the point P to have a destructive interference (or) be a dark fringe is,

Path difference, $\delta = (2n-1)\frac{\lambda}{2}$

where, $n = 1, 2, 3 \ldots$

$$\therefore \frac{d\,y}{D} = (2n-1)\frac{\lambda}{2}$$

$$y = \frac{(2n-1)}{2}\frac{\lambda D}{d} \text{ (or) } y_n = \frac{(2n-1)}{2}\frac{\lambda D}{d} \qquad (7.30)$$

UNIT-7(XII-Physics_Vol-2)_new.indd 64 21-07-2020 16:35:51